

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal
names in the project
```

```
## Clock signal
```

```
#set_property -dict { PACKAGE_PIN W5  IOSTANDARD LVCMOS33 } [get_ports clk]
#create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]
```

```
## Switches
```

```
set_property -dict { PACKAGE_PIN V17  IOSTANDARD LVCMOS33 } [get_ports {A[0]}]
set_property -dict { PACKAGE_PIN V16  IOSTANDARD LVCMOS33 } [get_ports {A[1]}]
set_property -dict { PACKAGE_PIN W16  IOSTANDARD LVCMOS33 } [get_ports {A[2]}]
set_property -dict { PACKAGE_PIN W17  IOSTANDARD LVCMOS33 } [get_ports {A[3]}]
set_property -dict { PACKAGE_PIN W15  IOSTANDARD LVCMOS33 } [get_ports {B[0]}]
set_property -dict { PACKAGE_PIN V15  IOSTANDARD LVCMOS33 } [get_ports {B[1]}]
set_property -dict { PACKAGE_PIN W14  IOSTANDARD LVCMOS33 } [get_ports {B[2]}]
set_property -dict { PACKAGE_PIN W13  IOSTANDARD LVCMOS33 } [get_ports {B[3]}]
set_property -dict { PACKAGE_PIN V2   IOSTANDARD LVCMOS33 } [get_ports {S}]
set_property -dict { PACKAGE_PIN T3   IOSTANDARD LVCMOS33 } [get_ports {CIN}]
#set_property -dict { PACKAGE_PIN T2   IOSTANDARD LVCMOS33 } [get_ports {CIN}]
#set_property -dict { PACKAGE_PIN T2   IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3   IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2   IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1   IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1   IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2   IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]
```

```
## LEDs
```

```
#set_property -dict { PACKAGE_PIN L1   IOSTANDARD LVCMOS33 } [get_ports {led[0]}]
#set_property -dict { PACKAGE_PIN E19  IOSTANDARD LVCMOS33 } [get_ports {led[1]}]
#set_property -dict { PACKAGE_PIN U19  IOSTANDARD LVCMOS33 } [get_ports {led[2]}]
#set_property -dict { PACKAGE_PIN V19  IOSTANDARD LVCMOS33 } [get_ports {led[3]}]
#set_property -dict { PACKAGE_PIN W18  IOSTANDARD LVCMOS33 } [get_ports {led[4]}]
#set_property -dict { PACKAGE_PIN U15  IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14  IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14  IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
#set_property -dict { PACKAGE_PIN V13  IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3   IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
#set_property -dict { PACKAGE_PIN W3   IOSTANDARD LVCMOS33 } [get_ports {led[10]}]
#set_property -dict { PACKAGE_PIN U3   IOSTANDARD LVCMOS33 } [get_ports {led[11]}]
```

```
#set_property -dict { PACKAGE_PIN P3   IOSTANDARD LVCMOS33 } [get_ports {led[12]]}
#set_property -dict { PACKAGE_PIN N3   IOSTANDARD LVCMOS33 } [get_ports {led[13]]}
#set_property -dict { PACKAGE_PIN P1   IOSTANDARD LVCMOS33 } [get_ports {led[14]]}
#set_property -dict { PACKAGE_PIN L1   IOSTANDARD LVCMOS33 } [get_ports {led[15]]}
```

##7 Segment Display

```
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {CA[6]]}
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {CA[5]]}
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {CA[4]]}
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {CA[3]]}
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {CA[2]]}
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {CA[1]]}
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {CA[0]]}
```

```
#set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports dp]
```

```
#set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports {an[0]]}
#set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports {an[1]]}
#set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports {an[2]]}
#set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports {an[3]]}
```

###Buttons

```
#set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports btnC]
#set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports btnU]
#set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports btnL]
#set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports btnR]
#set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports btnD]
```

##Pmod Header JA

```
#set_property -dict { PACKAGE_PIN J1   IOSTANDARD LVCMOS33 } [get_ports {JA[0]};#Sch
name = JA1
#set_property -dict { PACKAGE_PIN L2   IOSTANDARD LVCMOS33 } [get_ports {JA[1]};#Sch
name = JA2
#set_property -dict { PACKAGE_PIN J2   IOSTANDARD LVCMOS33 } [get_ports {JA[2]};#Sch
name = JA3
#set_property -dict { PACKAGE_PIN G2   IOSTANDARD LVCMOS33 } [get_ports {JA[3]};#Sch
name = JA4
#set_property -dict { PACKAGE_PIN H1   IOSTANDARD LVCMOS33 } [get_ports {JA[4]};#Sch
name = JA7
#set_property -dict { PACKAGE_PIN K2   IOSTANDARD LVCMOS33 } [get_ports {JA[5]};#Sch
name = JA8
```

```
#set_property -dict { PACKAGE_PIN H2  IOSTANDARD LVCMOS33 } [get_ports {JA[6]}];#Sch
name = JA9
#set_property -dict { PACKAGE_PIN G3  IOSTANDARD LVCMOS33 } [get_ports {JA[7]}];#Sch
name = JA10
```

##Pmod Header JB

```
#set_property -dict { PACKAGE_PIN A14  IOSTANDARD LVCMOS33 } [get_ports {JB[0]}];#Sch
name = JB1
#set_property -dict { PACKAGE_PIN A16  IOSTANDARD LVCMOS33 } [get_ports {JB[1]}];#Sch
name = JB2
#set_property -dict { PACKAGE_PIN B15  IOSTANDARD LVCMOS33 } [get_ports {JB[2]}];#Sch
name = JB3
#set_property -dict { PACKAGE_PIN B16  IOSTANDARD LVCMOS33 } [get_ports {JB[3]}];#Sch
name = JB4
#set_property -dict { PACKAGE_PIN A15  IOSTANDARD LVCMOS33 } [get_ports {JB[4]}];#Sch
name = JB7
#set_property -dict { PACKAGE_PIN A17  IOSTANDARD LVCMOS33 } [get_ports {JB[5]}];#Sch
name = JB8
#set_property -dict { PACKAGE_PIN C15  IOSTANDARD LVCMOS33 } [get_ports {JB[6]}];#Sch
name = JB9
#set_property -dict { PACKAGE_PIN C16  IOSTANDARD LVCMOS33 } [get_ports {JB[7]}];#Sch
name = JB10
```

##Pmod Header JC

```
#set_property -dict { PACKAGE_PIN K17  IOSTANDARD LVCMOS33 } [get_ports {JC[0]}];#Sch
name = JC1
#set_property -dict { PACKAGE_PIN M18  IOSTANDARD LVCMOS33 } [get_ports
{JC[1]}];#Sch name = JC2
#set_property -dict { PACKAGE_PIN N17  IOSTANDARD LVCMOS33 } [get_ports {JC[2]}];#Sch
name = JC3
#set_property -dict { PACKAGE_PIN P18  IOSTANDARD LVCMOS33 } [get_ports {JC[3]}];#Sch
name = JC4
#set_property -dict { PACKAGE_PIN L17  IOSTANDARD LVCMOS33 } [get_ports {JC[4]}];#Sch
name = JC7
#set_property -dict { PACKAGE_PIN M19  IOSTANDARD LVCMOS33 } [get_ports
{JC[5]}];#Sch name = JC8
#set_property -dict { PACKAGE_PIN P17  IOSTANDARD LVCMOS33 } [get_ports {JC[6]}];#Sch
name = JC9
#set_property -dict { PACKAGE_PIN R18  IOSTANDARD LVCMOS33 } [get_ports {JC[7]}];#Sch
name = JC10
```

##Pmod Header JXADC

```
#set_property -dict { PACKAGE_PIN J3  IOSTANDARD LVCMOS33 } [get_ports
{JXADC[0]}];#Sch name = XA1_P
```

```
#set_property -dict { PACKAGE_PIN L3  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[1]};#Sch name = XA2_P  
#set_property -dict { PACKAGE_PIN M2  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[2]};#Sch name = XA3_P  
#set_property -dict { PACKAGE_PIN N2  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[3]};#Sch name = XA4_P  
#set_property -dict { PACKAGE_PIN K3  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[4]};#Sch name = XA1_N  
#set_property -dict { PACKAGE_PIN M3  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[5]};#Sch name = XA2_N  
#set_property -dict { PACKAGE_PIN M1  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[6]};#Sch name = XA3_N  
#set_property -dict { PACKAGE_PIN N1  IOSTANDARD LVCMOS33 } [get_ports  
{JXADC[7]};#Sch name = XA4_N
```

##VGA Connector

```
#set_property -dict { PACKAGE_PIN G19  IOSTANDARD LVCMOS33 } [get_ports {vgaRed[0]}]  
#set_property -dict { PACKAGE_PIN H19  IOSTANDARD LVCMOS33 } [get_ports {vgaRed[1]}]  
#set_property -dict { PACKAGE_PIN J19  IOSTANDARD LVCMOS33 } [get_ports {vgaRed[2]}]  
#set_property -dict { PACKAGE_PIN N19  IOSTANDARD LVCMOS33 } [get_ports {vgaRed[3]}]  
#set_property -dict { PACKAGE_PIN N18  IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[0]}]  
#set_property -dict { PACKAGE_PIN L18  IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[1]}]  
#set_property -dict { PACKAGE_PIN K18  IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[2]}]  
#set_property -dict { PACKAGE_PIN J18  IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[3]}]  
#set_property -dict { PACKAGE_PIN J17  IOSTANDARD LVCMOS33 } [get_ports  
{vgaGreen[0]}]  
#set_property -dict { PACKAGE_PIN H17  IOSTANDARD LVCMOS33 } [get_ports  
{vgaGreen[1]}]  
#set_property -dict { PACKAGE_PIN G17  IOSTANDARD LVCMOS33 } [get_ports  
{vgaGreen[2]}]  
#set_property -dict { PACKAGE_PIN D17  IOSTANDARD LVCMOS33 } [get_ports  
{vgaGreen[3]}]  
#set_property -dict { PACKAGE_PIN P19  IOSTANDARD LVCMOS33 } [get_ports Hsync]  
#set_property -dict { PACKAGE_PIN R19  IOSTANDARD LVCMOS33 } [get_ports Vsync]
```

##USB-RS232 Interface

```
#set_property -dict { PACKAGE_PIN B18  IOSTANDARD LVCMOS33 } [get_ports RsRx]  
#set_property -dict { PACKAGE_PIN A18  IOSTANDARD LVCMOS33 } [get_ports RsTx]
```

##USB HID (PS/2)

```
#set_property -dict { PACKAGE_PIN C17  IOSTANDARD LVCMOS33  PULLUP true }  
[get_ports PS2Clk]  
#set_property -dict { PACKAGE_PIN B17  IOSTANDARD LVCMOS33  PULLUP true }  
[get_ports PS2Data]
```

##Quad SPI Flash

##Note that CCLK_0 cannot be placed in 7 series devices. You can access it using the
##STARTUPE2 primitive.

```
#set_property -dict { PACKAGE_PIN D18  IOSTANDARD LVCMOS33 } [get_ports {QspiDB[0]}]  
#set_property -dict { PACKAGE_PIN D19  IOSTANDARD LVCMOS33 } [get_ports {QspiDB[1]}]  
#set_property -dict { PACKAGE_PIN G18  IOSTANDARD LVCMOS33 } [get_ports {QspiDB[2]}]  
#set_property -dict { PACKAGE_PIN F18  IOSTANDARD LVCMOS33 } [get_ports {QspiDB[3]}]  
#set_property -dict { PACKAGE_PIN K19  IOSTANDARD LVCMOS33 } [get_ports QspiCSn]
```

Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]